

Week 5: MEC6602E Transonic Aerodynamics

Frédéric Plante, ing., Ph.D.

Polytechnique Montréal

September 29, 2026



Thermodynamics

Adiabatic: No heat exchange with exterior of control volume.

Isentropic: Reversible and adiabatic process (Shock waves are adiabatic, but not reversible).

Perfect gas:

$$p = \rho RT$$

First law:

$$\delta q + \delta w = de$$

$$T ds = de + p dv$$

$$T ds = dh - v dp$$

Second law:

$$ds \geq \frac{\delta q}{T}$$

Calorically perfect gas

$$s_2 - s_1 = c_p \ln \frac{T_2}{T_1} - R \ln \frac{p_2}{p_1}$$

$$s_2 - s_1 = c_v \ln \frac{T_2}{T_1} + R \ln \frac{v_2}{v_1}$$

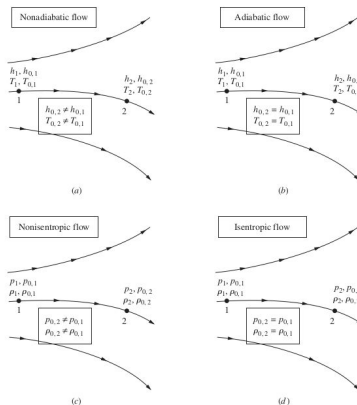


Figure 7.4 Comparisons between (a) nonadiabatic, (b) adiabatic, (c) nonisentropic, and (d) isentropic flows.

Isentropic relation

$$\frac{p_2}{p_1} = \left(\frac{\rho_2}{\rho_1}\right)^\gamma = \left(\frac{T_2}{T_1}\right)^{\gamma/(\gamma-1)}$$

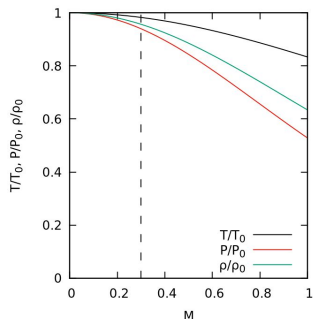
$$\frac{p}{p_t} = \left(1 + \frac{\gamma-1}{2} M^2\right)^{-\frac{\gamma}{\gamma-1}}$$

$$\frac{T}{T_t} = \left(1 + \frac{\gamma-1}{2} M^2\right)^{-1}$$

$$\frac{\rho}{\rho_t} = \left(1 + \frac{\gamma-1}{2} M^2\right)^{-\frac{1}{\gamma-1}}$$

Tabulated

M	$\frac{p_0}{p}$	$\frac{\rho_0}{\rho}$	$\frac{T_0}{T}$
0.7100 + 01	0.4531 + 04	0.4088 + 03	0.1108 + 02
0.7200 + 01	0.4953 + 04	0.4357 + 03	0.1137 + 02
0.7300 + 01	0.5410 + 04	0.4640 + 03	0.1166 + 02
0.7400 + 01	0.5903 + 04	0.4939 + 03	0.1195 + 02
0.7500 + 01	0.6434 + 04	0.5252 + 03	0.1225 + 02
0.7600 + 01	0.7006 + 04	0.5582 + 03	0.1255 + 02
0.7700 + 01	0.7623 + 04	0.5928 + 03	0.1286 + 02
0.7800 + 01	0.8285 + 04	0.6292 + 03	0.1317 + 02
0.7900 + 01	0.8998 + 04	0.6674 + 03	0.1348 + 02
0.8000 + 01	0.9763 + 04	0.7075 + 03	0.1380 + 02



Total temperature and enthalpy: Fluid is brought to rest adiabatically.

Total pressure and density: Fluid is brought to rest isentropically.

ANDERSON. (2016). *Fundamentals of aerodynamics* (6th ed.). McGraw-Hill Education.

Normal shock

Equations:

$$M_2^2 = \frac{1 + [(\gamma - 1)/2]M_1^2}{\gamma M_1^2 - (\gamma - 1)/2}$$

$$\frac{\rho_2}{\rho_1} = \frac{u_1}{u_2} = \frac{(\gamma + 1)M_1^2}{2 + (\gamma - 1)M_1^2}$$

$$\frac{p_2}{p_1} = 1 + \frac{2\gamma}{\gamma + 1}(M_1^2 - 1)$$

$$\frac{T_2}{T_1} = \frac{h_2}{h_1} = \left[1 + \frac{2\gamma}{\gamma + 1}(M_1^2 - 1)\right] \frac{2 + (\gamma - 1)M_1^2}{(\gamma + 1)M_1^2}$$

Properties downstream of the shock depend on the ones upstream of the shock. We often use tables.

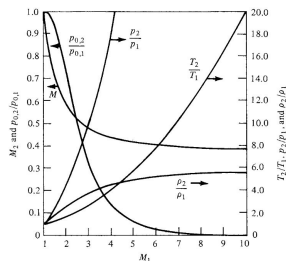


Figure 8.8 The variation of properties across a normal shock wave as a function of upstream Mach number: $\gamma = 1.4$.

M	$\frac{p_2}{p_1}$	$\frac{\rho_2}{\rho_1}$	$\frac{T_2}{T_1}$	$\frac{p_{0,2}}{p_{0,1}}$	$\frac{p_{0,2}}{p_1}$	M_2
0.1000 + 01	0.1000 + 01	0.1000 + 01	0.1000 + 01	0.1000 + 01	0.1893 + 01	0.1000 + 01
0.1020 + 01	0.1047 + 01	0.1033 + 01	0.1013 + 01	0.1000 + 01	0.1938 + 01	0.9805 + 00
0.1040 + 01	0.1095 + 01	0.1067 + 01	0.1026 + 01	0.9999 + 00	0.1984 + 01	0.9620 + 00
0.1060 + 01	0.1144 + 01	0.1101 + 01	0.1039 + 01	0.9998 + 00	0.2032 + 01	0.9444 + 00
0.1080 + 01	0.1194 + 01	0.1135 + 01	0.1052 + 01	0.9994 + 01	0.2082 + 01	0.9277 + 00
0.1100 + 01	0.1245 + 01	0.1169 + 01	0.1065 + 01	0.9989 + 00	0.2133 + 01	0.9118 + 00
0.1120 + 01	0.1297 + 01	0.1203 + 01	0.1078 + 01	0.9982 + 00	0.2185 + 01	0.8966 + 00
0.1140 + 01	0.1350 + 01	0.1238 + 01	0.1090 + 01	0.9973 + 00	0.2239 + 01	0.8820 + 00
0.1160 + 01	0.1403 + 01	0.1272 + 01	0.1103 + 01	0.9961 + 00	0.2294 + 01	0.8682 + 00
0.1180 + 01	0.1458 + 01	0.1307 + 01	0.1115 + 01	0.9946 + 00	0.2350 + 01	0.8549 + 00

ANDERSON. (2016). *Fundamentals of aerodynamics* (6th ed.). McGraw-Hill Education.

Subsonic nozzle

- ① If $P_e = P_0$, no flow;
- ② If $P_{e,3} \leq P_e \leq P_0$, flow is always subsonic;
- ③ If $P_e = P_{e,3}$, flow is sonic at the throat but subsonic downstream. The ratio P/P_0 at the throat is 0.528.

There is an infinity of subsonic solutions, but the maximal mass flow is when $P_e = P_{e,3}$. This is because the Mach number cannot be more than 1 at the throat.

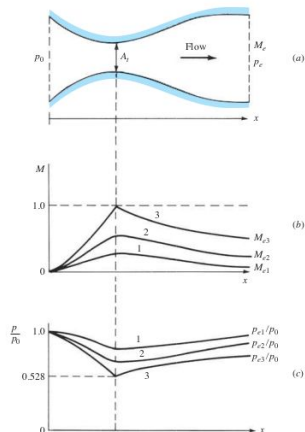


Figure 10.13 Isentropic subsonic nozzle flow.

ANDERSON. (2016). *Fundamentals of aerodynamics* (6th ed.). McGraw-Hill Education.

Choked flow

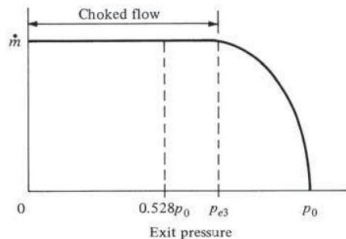


Figure 10.14 Variation of mass flow with exit pressure; illustration of choked flow.

For every back pressure lower than $P_{e,3}$, the mass flow is constant.

If $P_e < P_{e,3}$, the outflow is supersonic.

The flow in the convergent part (upstream of the throat) is independent of the back pressure in this case. Information does not travel upstream in a supersonic flow.

ANDERSON. (2016). *Fundamentals of aerodynamics* (6th ed.). McGraw-Hill Education.



Isentropic supersonic nozzle

If the flow expands isentropically:

- 1 The Mach distribution is prescribed by the shape of the nozzle (A/A^*)
- 2 We can compute the pressure at any point

If $P_e = P_{e,6}$ there is only one possible solution and it is isentropic. Otherwise, the flow is not isentropic.

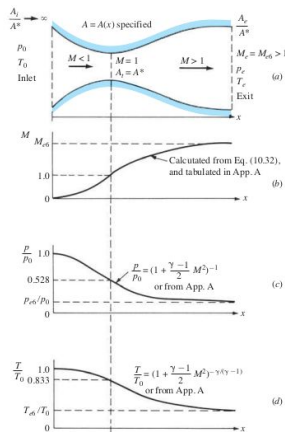


Figure 10.12 Isentropic supersonic nozzle flow.

ANDERSON. (2016). *Fundamentals of aerodynamics* (6th ed.). McGraw-Hill Education.

What happens if $P_{e,6} < P_e < P_{e,3}$?

There will be a normal shock wave;

After a normal shock wave, the flow is subsonic;

A divergent nozzle slows down a subsonic flow, so the outflow is subsonic.

The position of the shock is prescribed by the back pressure.

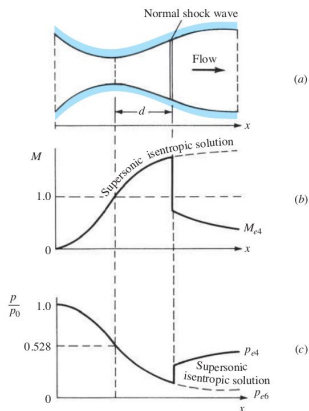
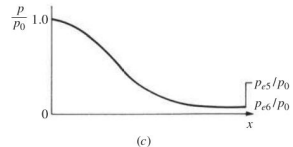
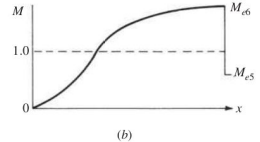
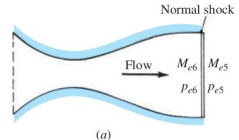


Figure 10.15 Supersonic nozzle flow with a normal shock inside the nozzle.

If $P_{e,6} < P_e < P_{e,3}$ _____

The shock can be right at the end of the nozzle.



ANDERSON. (2016). *Fundamentals of aerodynamics* (6th ed.). McGraw-Hill Education.

Effect of Pressure downstream of the outlet

- If subsonic: $P_e = P_B$
- If $P_B = P_{e,6}$: No shock wave
- If $P_{e,6} < P_B < P_{e,5}$:
Over-expanded, there are oblique shocks at the exit to compress the gas.
- If $P_B < P_{e,6}$:
Under-expanded, there are expansion waves to decompress the gas isentropically.

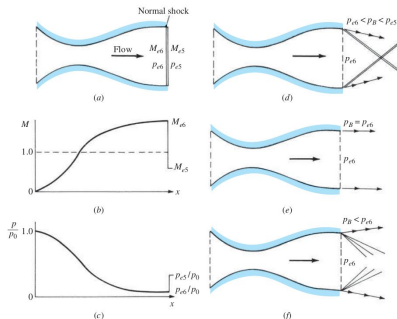


Figure 10.16 Supersonic nozzle flows with waves at the nozzle exit: (a), (b), and (c) pertain to a normal shock at the exit, (d) overexpanded nozzle, (e) isentropic expansion to the back pressure equal to the exit pressure, (f) underexpanded nozzle.

ANDERSON. (2016). *Fundamentals of aerodynamics* (6th ed.). McGraw-Hill Education.

Tuyère



'underexpanded'



'overexpanded'



'overexpanded'

Images internet



Quasi-1D Mac Cormack Method

$$Q = \begin{bmatrix} \rho A \\ \rho u A \\ e A \end{bmatrix}, \quad E = \begin{bmatrix} \rho u A \\ (\rho u^2 + p) A \\ u(e + p) A \end{bmatrix} \quad \text{and} \quad S = \begin{bmatrix} 0 \\ p \frac{dA}{dx} \\ 0 \end{bmatrix}$$

predictor step:

$$\bar{Q}_i = Q_i^n - \Delta t \left(\frac{E_i^n - E_{i-1}^n}{\Delta x} \right) + \Delta t S_i^n$$

corrector step:

$$\bar{\bar{Q}}_i = Q_i^n - \Delta t \left(\frac{\bar{E}_{i+1} - \bar{E}_i}{\Delta x} \right) + \Delta t \bar{S}_i^n$$



Lax-Wendroff scheme

Taylor expansion

$$Q^{n+1} = Q^n + \frac{\partial Q}{\partial t} \Delta t + \frac{\partial^2 Q}{\partial t^2} \frac{\Delta t^2}{2}$$

Express the time derivative as space derivative

$$\frac{\partial Q}{\partial t} = -\frac{\partial E}{\partial x}$$

$$\frac{\partial^2 Q}{\partial t^2} = -\frac{\partial}{\partial x} \left(\frac{\partial E}{\partial t} \right) = -\frac{\partial}{\partial x} \left(\frac{\partial E}{\partial Q} \frac{\partial Q}{\partial t} \right) = -\frac{\partial}{\partial x} \left(A \frac{\partial Q}{\partial t} \right) = \frac{\partial}{\partial x} \left(A \frac{\partial E}{\partial x} \right)$$

$$Q^{n+1} = Q^n + \frac{\partial Q}{\partial t} \Delta t + \frac{\partial}{\partial x} \left(A \frac{\partial E}{\partial x} \right) \frac{\Delta t^2}{2}$$

Discretize with central differences

$$Q^{n+1} = Q^n + \frac{\Delta t}{\Delta x} \left(\frac{E_{i+1}^n - E_{i-1}^n}{2} \right) +$$

$$\frac{\Delta t^2}{2\Delta x^2} \left[A_{i+1/2}^n (E_{i+1}^n - E_i^n) - A_{i-1/2}^n (E_i^n - E_{i-1}^n) \right]$$

Compute the Jacobian at the middle coordinate $A_{i+1/2} = \frac{A_i + A_{i+1}}{2}$



Implicit Beam-Warming Method

$$\frac{\partial Q}{\partial t} + \frac{\partial E}{\partial x} = 0$$

$$\left[I + \Delta t \delta_x A^n - \varepsilon_i (\nabla \Delta)_x \right] \Delta Q^n = -\Delta t \delta_x E - \varepsilon_e (\nabla \Delta)_x^2 Q^n$$

$$\varepsilon_i > 2\varepsilon_e$$

$$\varepsilon_e < 0.125$$

Source: Cebeci, T., Shao, J.P., Kafyeke, F., Laurendeau, E., *Computational Fluid Dynamics for Engineers*, Springer.

The term in $[]$ is a block tridiagonal matrix. $\delta_x E = \frac{E_{i+1} - E_{i-1}}{\Delta x}$ is the central operator. Same thing apply to the jacobian matrices in the left and side. To get the solution change ΔQ we solve the linear system.



Implicit Beam-Warming Method

Fourth order operator: $(\nabla\Delta)_x^2 u = \frac{u_{i-2} - 4u_{i-1} + 6u_i - 4u_{i+1} + u_{i+2}}{\Delta x^4}$

Second order operator: $(\nabla\Delta)_x u = \frac{u_{i-1} - 2u_i + u_{i+1}}{\Delta x^2}$

- Using the second order operator, the LHS matrix is bloc tridiagonal and easily invertible.
- In the RHS, the fourth or the second order operator can be used (but the sign change!!)
- In practice we lump the Δx^2 or Δx^4 division in the constant ϵ_i and ϵ_e
- The constant ϵ_i and ϵ_e are multiplied by Δt to be consistent
- The Second or Fourth order dissipation term can be used



Solving a block tridiagonal system

We have a linear system of the form (with 3×3 blocks)

$$L_i \Delta Q_{i-1} + D_i \Delta Q_i + U_i \Delta Q_{i+1} = R_i$$

The Thomas algorithm carries over. First sweep:

$$C'_i = \begin{cases} D_1^{-1} U_1 & i = 1 \\ (D_i - L_i C'_{i-1})^{-1} U_i & i = 2, \dots, n-1 \end{cases}$$

$$R'_i = \begin{cases} D_1^{-1} R_1 & i = 1 \\ (D_i - L_i C'_{i-1})^{-1} (R_i - L_i R'_{i-1}) & i = 2, \dots, n \end{cases}$$

Backward substitution, with no inverse left to compute:

$$\Delta Q_n = R'_n, \quad \Delta Q_i = R'_i - C'_i \Delta Q_{i+1} \quad i = n-1, \dots, 1$$

In this algorithm we only need to invert 3×3 matrices.



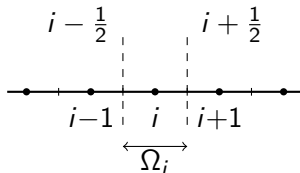
Finite volume vs finite difference

FD works on the *differential* form;

FV works on the *integral* form.

Integrate $u_t + cu_x = 0$ over a control volume Ω_i and apply the divergence theorem:

$$\Omega_i \frac{\partial u_i}{\partial t} + (f_{i+1/2} - f_{i-1/2}) = 0, \quad \Omega_i = \Delta x, \quad f = cu.$$



General FV scheme (any face flux):

$$u_i^{n+1} = u_i^n - \frac{\Delta t}{\Delta x} (f_{i+1/2} - f_{i-1/2})$$

All FV schemes share this form; they differ only in how $f_{i\pm 1/2}$ is built.

Key point. Taking $f_{i\pm 1/2}$ as the average of the two neighbouring cells gives

$u_i^{n+1} = u_i^n - \frac{c\Delta t}{2\Delta x} (u_{i+1} - u_{i-1})$: the central FTCS scheme, unstable. **On a uniform 1D grid, FV with a central flux is exactly FD.**

Stabilizing the flux: upwind and Roe

Central fluxes are unstable; add dissipation directly in the face flux (Roe-type flux), with $a = \partial f / \partial u$ the flux Jacobian:

$$f_{i+1/2} = \frac{1}{2} \left[f_{i+1} + f_i - |a|_{i+1/2} (u_{i+1} - u_i) \right]$$

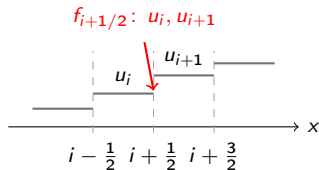
For $f = cu$, $|a|_{i+1/2} = |c|$. If $c > 0$: $f_{i+1/2} = c u_i$, $f_{i-1/2} = c u_{i-1}$, so

$$u_i^{n+1} = u_i^n - \frac{c \Delta t}{\Delta x} (u_i^n - u_{i-1}^n) \quad (\text{the upwind scheme}).$$

Key point. Every numerical flux is a central part plus a dissipation:

$$f_{i+1/2} = \underbrace{\frac{1}{2}(f_i + f_{i+1})}_{\text{central}} - \underbrace{\frac{1}{2}|a|_{i+1/2}(u_{i+1} - u_i)}_{\text{dissipation}}$$

Lax–Wendroff written in FV form (face flux with the $A_{i\pm 1/2}$ correction) has this same structure, and so does the Roe flux for the Euler system, with $|a| \rightarrow |A| = R|\Lambda|R^{-1}$ at a Roe-averaged state $\hat{Q}$.



Higher order: the MUSCL method

The upwind flux uses a *piecewise-constant* state: u_i left of face $i + \frac{1}{2}$, u_{i+1} right of it. MUSCL keeps the same update and upwind flux, but reconstructs a *piecewise-linear* state in each cell, giving two distinct face values u^L, u^R :

$$u_{i+1/2}^L = u_i + \frac{1}{2}\psi(r_i)(u_i - u_{i-1})$$

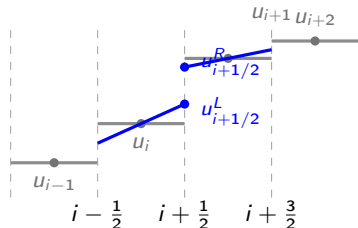
$$r_i = \frac{u_{i+1} - u_i}{u_i - u_{i-1}}$$

$$u_{i+1/2}^R = u_{i+1} - \frac{1}{2}\psi(r_{i+1})(u_{i+2} - u_{i+1})$$

Flux fed with the two states instead of u_i, u_{i+1} :

$$f_{i+1/2} = \frac{1}{2} \left[c(u_{i+1/2}^L + u_{i+1/2}^R) - |c|(u_{i+1/2}^R - u_{i+1/2}^L) \right]$$

$\psi(r) = 0$ recovers the first-order upwind scheme.

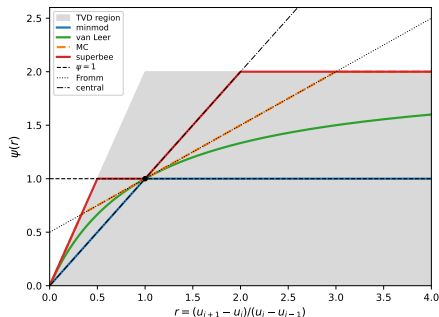


The limiter $\psi(r)$

limiter	$\psi(r)$
minmod	$\max(0, \min(1, r))$
van Leer	$\frac{r + r }{1 + r }$
MC	$\max(0, \min(2r, \frac{1+r}{2}, 2))$
superbee	$\max(0, \min(2r, 1), \min(r, 2))$

$r_i = \frac{u_{i+1} - u_i}{u_i - u_{i-1}}$ compares the downstream and upstream gradients of cell

- $r \simeq 1$ is smooth
- $r \leq 0$ is a local extremum
- r far from 1 signals a discontinuity



minmod: most diffusive superbee: most compressive